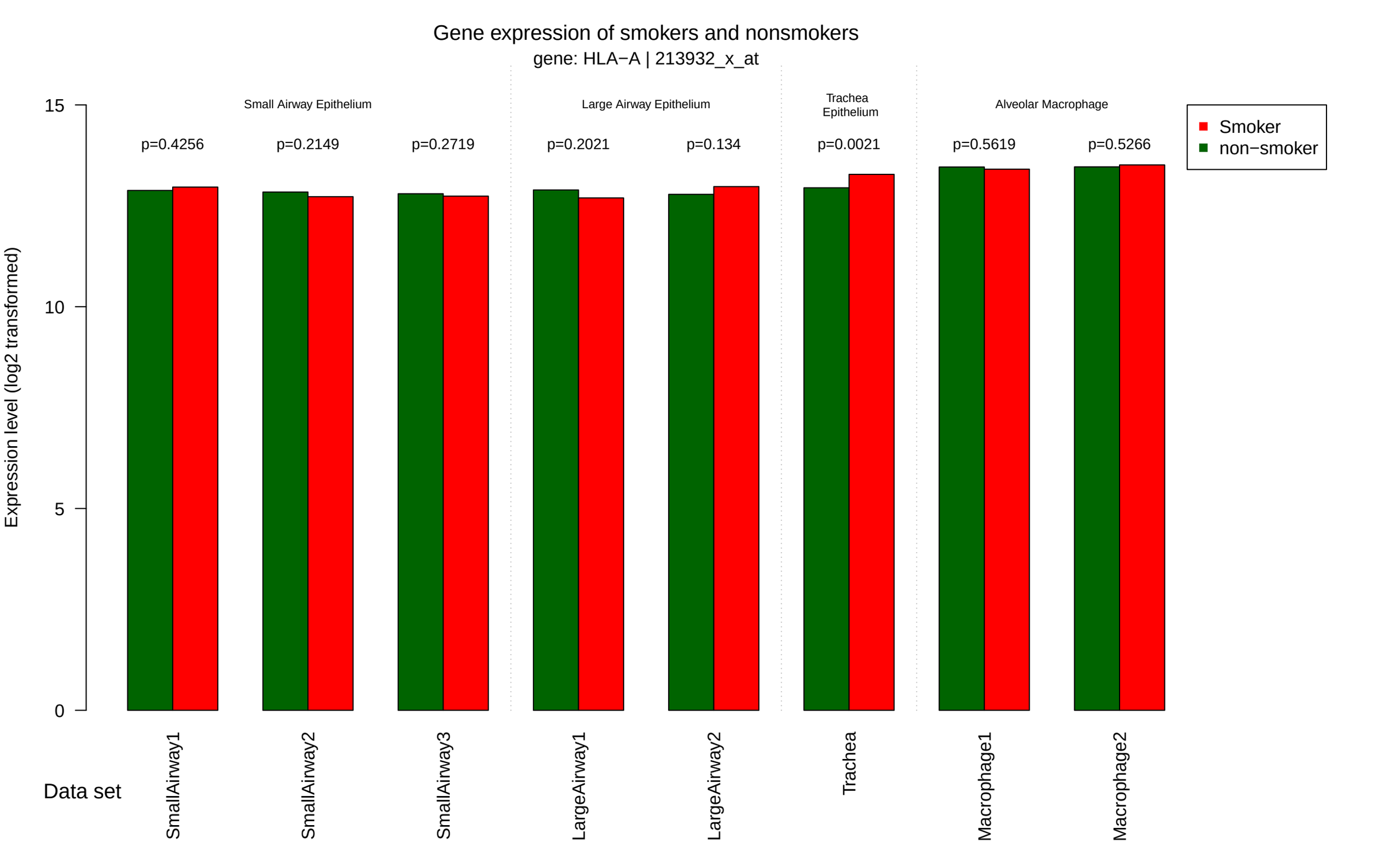
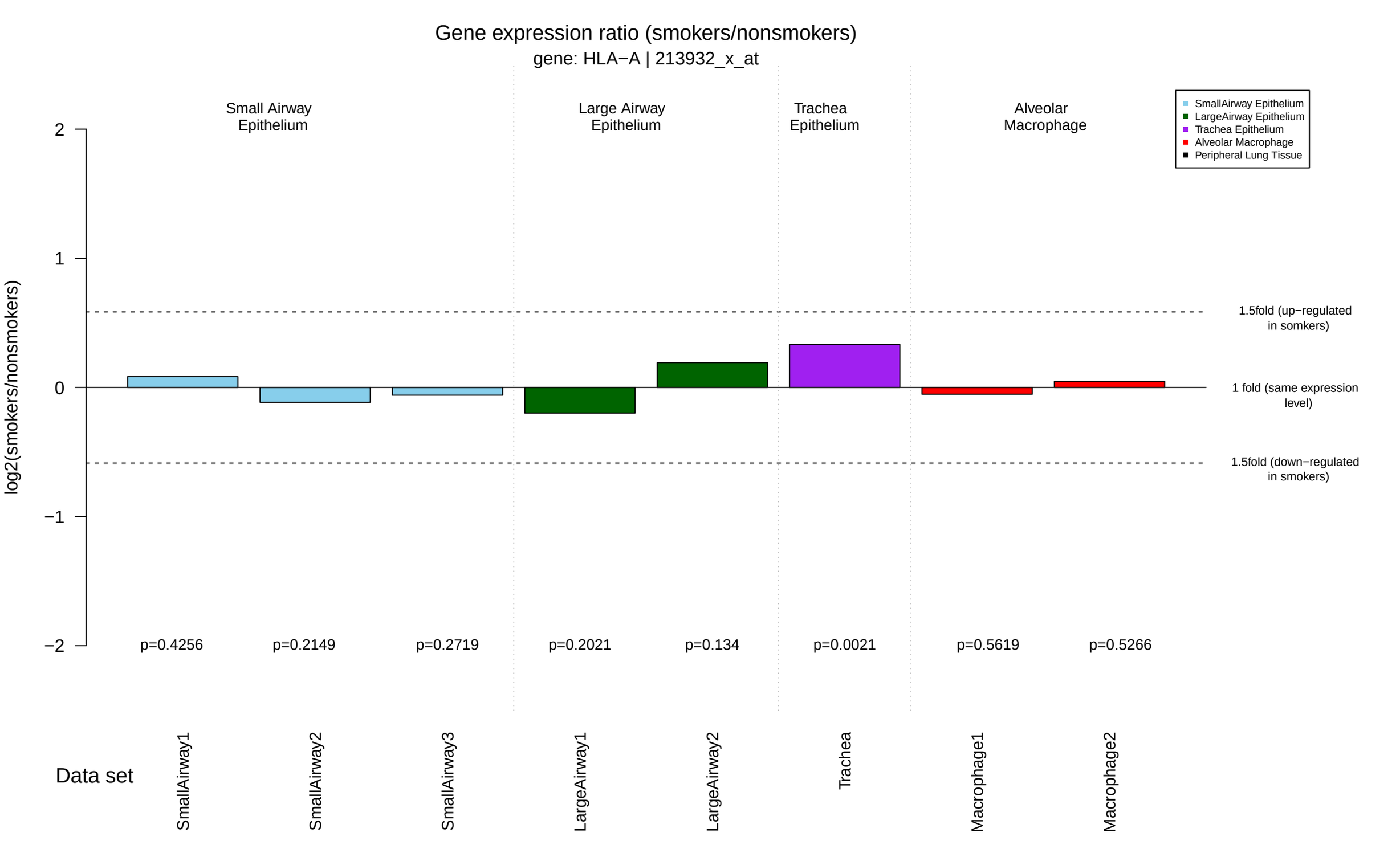
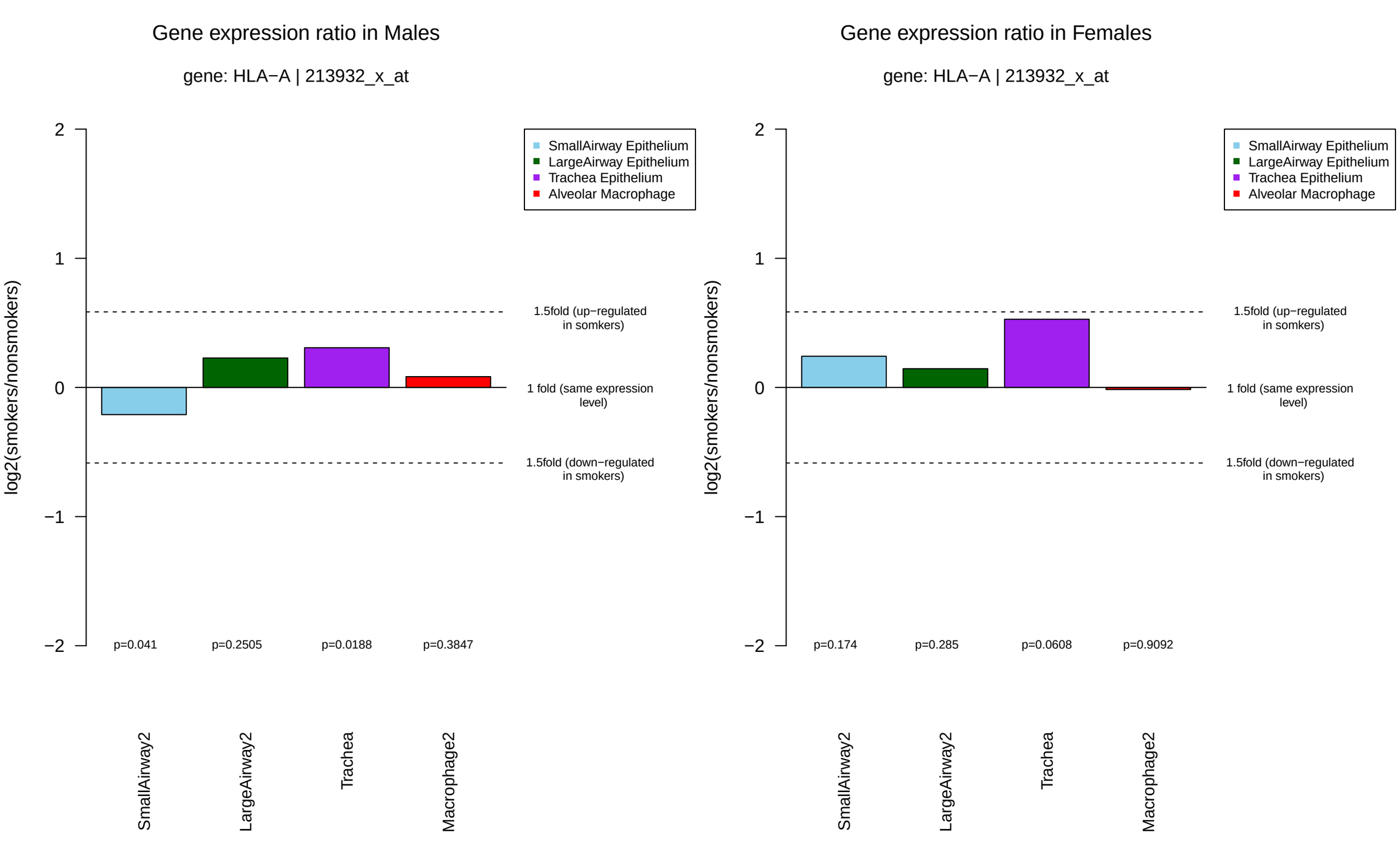
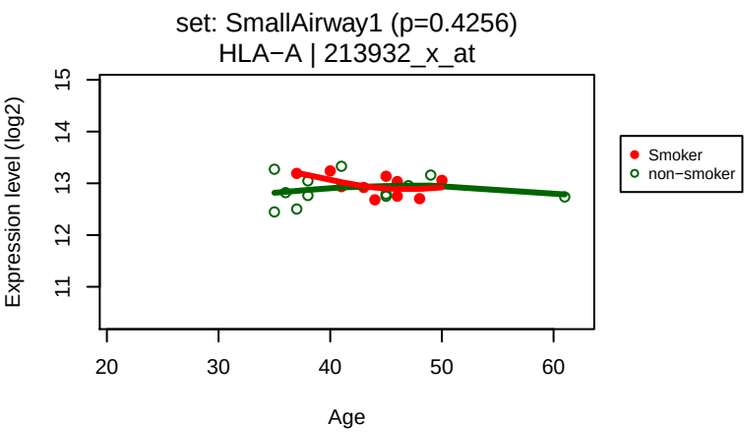
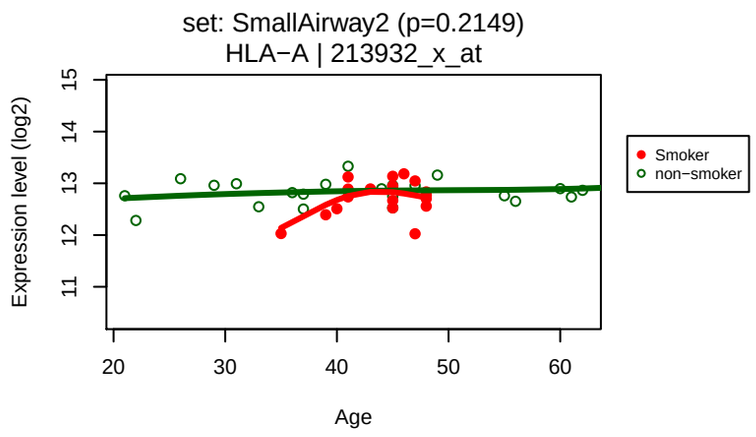
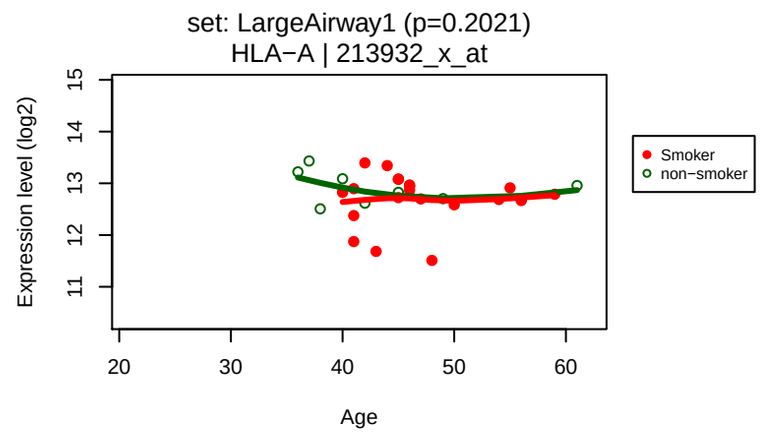

Gene expression of smokers and nonsmokers
gene: HLA−A | 213932_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4256
p=0.2149
p=0.2719
p=0.2021
p=0.134
p=0.0021
p=0.5619
p=0.5266
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

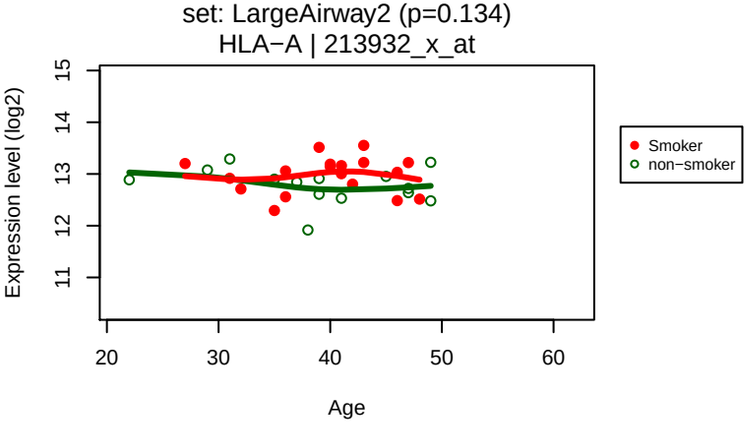

Gene expression ratio (smokers/nonsmokers)
gene: HLA−A | 213932_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4256
p=0.2149
p=0.2719
p=0.2021
p=0.134
p=0.0021
p=0.5619
p=0.5266
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

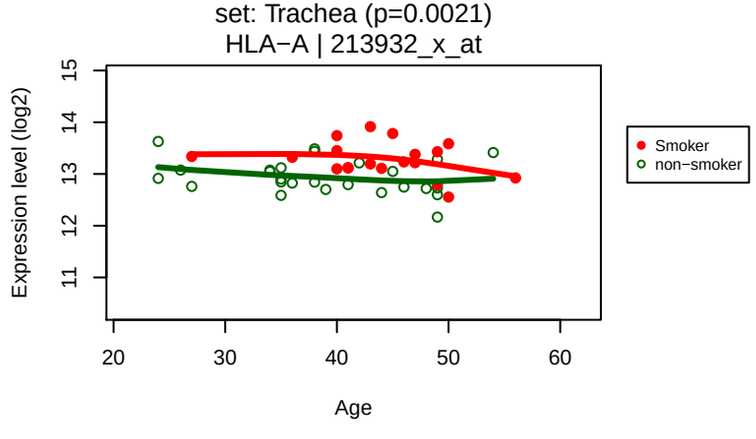

Gene expression ratio in Males
gene: HLA−A | 213932_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.041
p=0.2505
p=0.0188
p=0.3847
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−A | 213932_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.174
p=0.285
p=0.0608
p=0.9092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

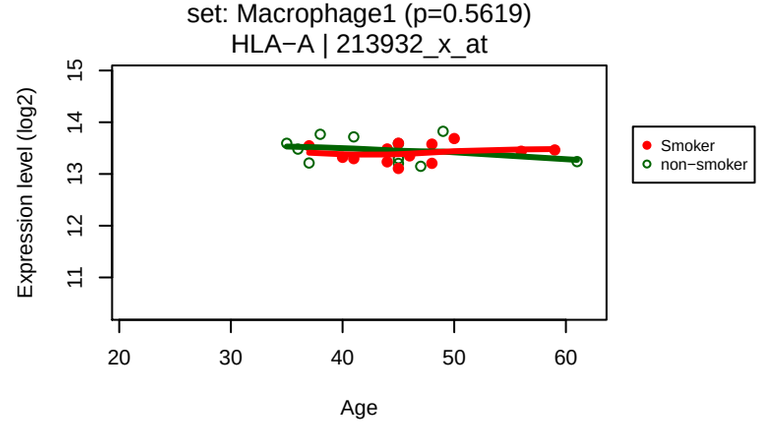

set: SmallAirway1 (p=0.4256)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker

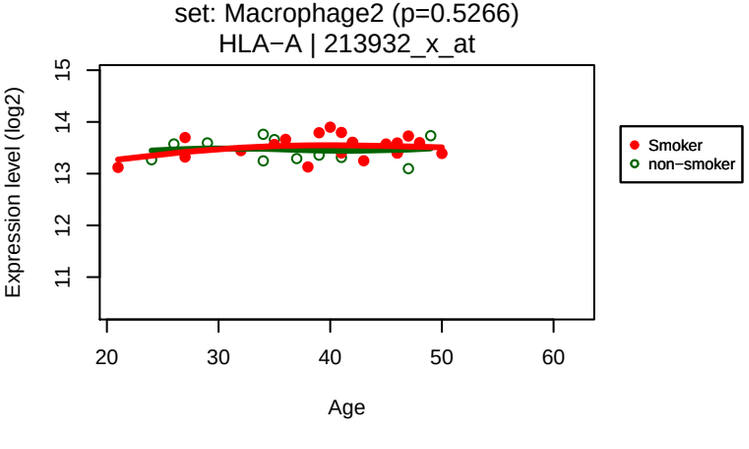

set: SmallAirway2 (p=0.2149)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker

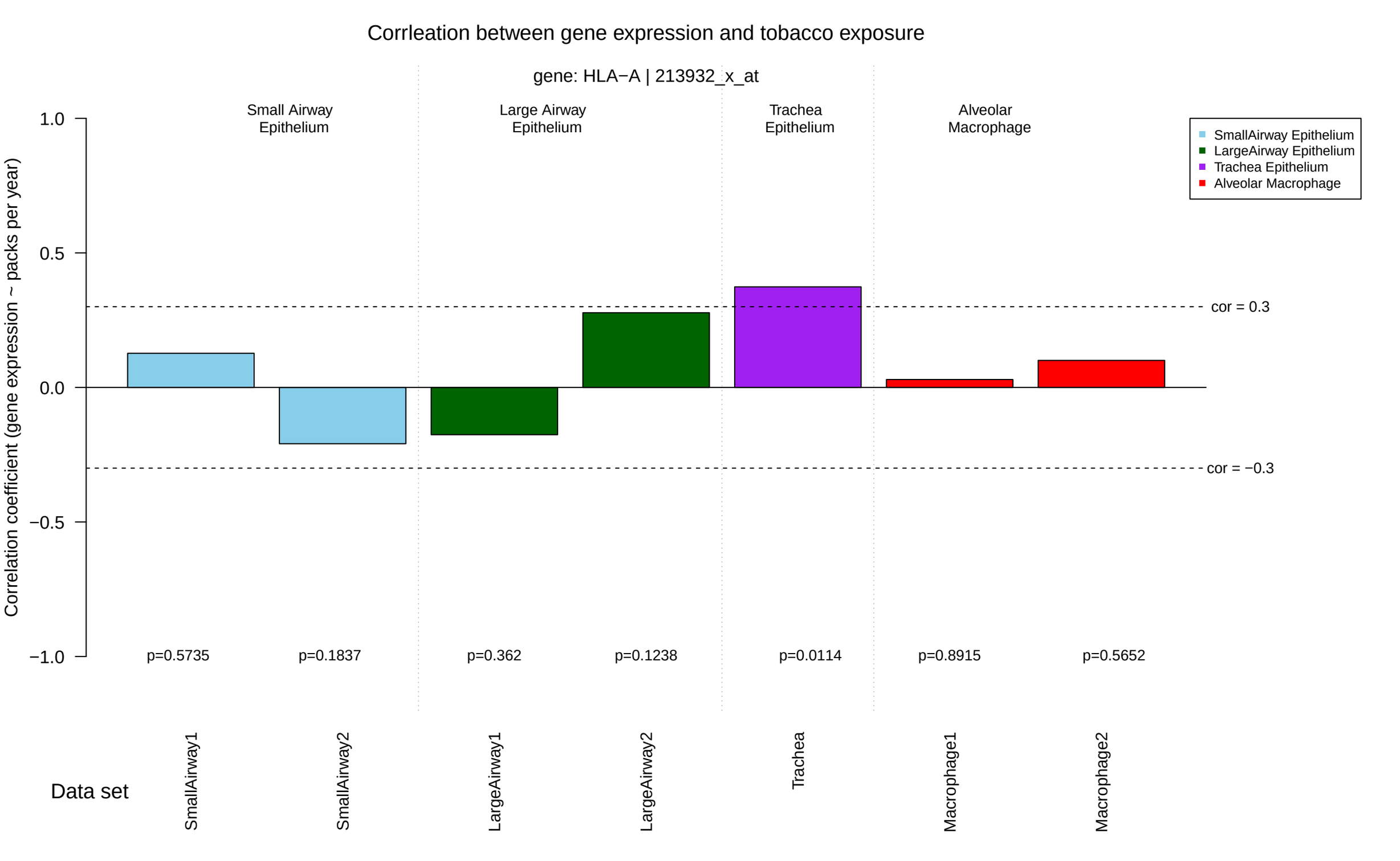

set: LargeAirway1 (p=0.2021)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.134)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0021)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5619)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5266)
HLA−A | 213932_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−A | 213932_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5735
p=0.1837
p=0.362
p=0.1238
p=0.0114
p=0.8915
p=0.5652
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2